NutriAI: AI-Based Sri Lankan Food Recognition and Budget-Aware Nutrition Recommendation System for the Aegis Life Platform

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Project Proposal Report

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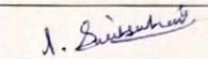
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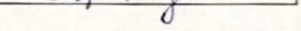
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

In recent years, digital health technologies have helped people monitor and manage their health more easily. Nutrition management systems are widely used to track calorie intake, understand eating habits, and support healthier lifestyles. However, most existing nutrition applications are mainly designed using Western food databases and require users to manually enter their meals. This makes them less suitable for countries like Sri Lanka, where food culture is very different. Sri Lankan meals often include combinations such as rice with several curries, hoppers, and string hoppers, which are not properly represented in many global nutrition databases. Because of this, users often have to guess or approximate their food entries, which leads to inaccurate nutrition analysis.

Another problem is that manual food logging takes time and can easily lead to mistakes. This reduces user interest and makes it difficult for people to consistently use these applications. In addition, many existing systems do not consider the affordability of food when giving dietary recommendations, which makes them less practical for many users.

This research proposes **NutriAI**, an AI-powered Sri Lankan food recognition and nutrition management system that will be developed as part of the **Aegis Life digital wellness platform**. The system will use Convolutional Neural Networks (CNNs) to recognize Sri Lankan food items from images taken by users. Once the food is identified, the system will analyze it using a localized nutrition database to estimate calorie and nutrient intake. Based on the user's profile, health goals, and budget, the system will provide personalized and affordable nutrition recommendations.

The system will be developed using technologies such as Python, TensorFlow or PyTorch, OpenCV, RESTful APIs, and a MySQL or MongoDB database. The expected outcome is an intelligent nutrition management system that makes food tracking easier, improves nutrition accuracy, and encourages healthier eating habits among Sri Lankan users.

Overall, this system helps promote preventive healthcare and raises awareness about healthy eating. It can create a **positive social impact** by helping people build healthier habits and reduce the risk of diet-related diseases. In addition, it can have a **commercial impact** by creating opportunities to work with healthcare providers, nutrition experts, and digital wellness platforms, while supporting the development of useful health technology and affordable diet planning solutions.

Keywords: Artificial Intelligence, Food Image Recognition, Nutrition Management, Deep Learning, Sri Lankan Cuisine

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
CNN	Convolutional Neural Network
DL	Deep Learning
ML	Machine Learning
API	Application Programming Interface
RDA	Recommended Dietary Allowance
SDG	Sustainable Development Goals
UI	User Interface
UX	User Experience
DBMS	Database Management System
REST	Representational State Transfer
NCD	Non-Communicable Diseases

Table 1

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1. INTRODUCTION

1.1 Background and Context

Maintaining a healthy diet is very important for preventing many lifestyle diseases such as obesity, diabetes, and heart-related conditions. With the rapid growth of digital health technologies, many people now use mobile applications and online platforms to track their daily nutrition and maintain healthier lifestyles. Artificial Intelligence (AI) has become an important technology in healthcare because it can analyze large amounts of data, provide personalized recommendations, and support preventive healthcare management [1].

Several nutrition and diet-tracking applications such as MyFitnessPal, Lose It!, and HealthifyMe have been developed to help users record their meals, track calorie intake, and understand their eating habits. These applications provide features like calorie counting, food databases, and progress tracking. However, most of these systems still rely on manual food logging. Users must search for food items and estimate portion sizes by themselves. This process can take time and may lead to inaccurate data entry, which reduces the accuracy of nutrition analysis [2].

Some modern applications have started using image-based food recognition with deep learning technologies. Although this approach is promising, it still has several limitations. Many food recognition models are trained using global datasets such as Food-101 and UECFood256, which mainly contain Western or East Asian foods. Because of this, these models do not perform well when identifying traditional Sri Lankan foods, since local food images are not well represented in these datasets [5][6].

Another limitation is that many existing systems are designed for a global audience and do not consider local eating habits, ingredients, or food combinations. Sri Lankan meals often include multiple dishes served together, such as rice with several curries, hoppers, string hoppers, and kottu. These meals are not well represented in global food databases. As a result, many systems struggle to correctly recognize these foods or provide accurate nutritional information. In addition, most nutrition applications do not consider the cost of food when suggesting diet plans, which is an important factor for many users in developing countries.

To overcome these limitations, this research proposes NutriAI, an AI-based Sri Lankan food recognition and nutrition management system that will be integrated into the Aegis Life wellness platform. The system will use deep learning techniques to identify Sri Lankan foods from images and estimate their nutritional values using a localized food database. It will also provide personalized and budget-friendly diet recommendations to help users maintain healthier eating habits.

This research also supports the United Nations Sustainable Development Goal 3 (Good Health and Well-Being) by encouraging healthy eating habits and improving access to nutrition guidance.

1.2 Literature Review

Recent studies have explored how Artificial Intelligence (AI) and machine learning can be used in food recognition and nutrition recommendation systems. Many researchers use deep learning models such as Convolutional Neural Networks (CNNs) to identify food items from images. These models can learn visual patterns from food pictures and classify different types of meals. For example, the DeepFood model showed that CNN-based systems can recognize food images with high accuracy, making AI useful for automatic diet monitoring [1].

Researchers also use large food image datasets to train these models. Popular datasets such as Food-101 and UECFood256 contain thousands of labeled food images and allow systems to recognize many types of food [5][6]. These datasets have helped improve research in food recognition technology. However, they mainly contain Western or Japanese food items, which makes it difficult for these models to correctly identify foods from other cultures.

Many studies have also developed AI-based diet recommendation systems. These systems analyze user information such as age, weight, health conditions, and lifestyle to suggest personalized meal plans [2][9]. Some systems combine food recognition with nutrition analysis to give real-time feedback about calories and nutrients. This helps users understand their eating habits and make healthier choices.

Some research has also focused on creating affordable diet plans using optimization methods. These systems generate meal plans that meet nutritional requirements while keeping food costs low [8]. This approach is useful in countries where people need healthy diets that are also affordable.

There are also many commercial applications that help users track their diets and improve their health. Popular apps such as MyFitnessPal, Lose It!, and HealthifyMe allow users to log meals, count calories, and track their nutrition. Some apps also provide simple diet suggestions. However, most of these applications depend on manual food entry and global food databases, which may not accurately represent traditional local foods.

Although many technologies and applications exist, they still have some limitations. Most systems are designed for global users and do not include many traditional foods from developing countries. In addition, many applications do not consider the cost of meals when recommending diets. Because of these limitations, there is a need for a localized AI-based nutrition management system that can recognize Sri Lankan foods and provide practical and affordable diet recommendations.

1.3 Research Gap

Despite improvements in AI-based food recognition and nutrition systems, there are still several gaps. First, many food recognition models are trained using global datasets that do not include many Sri Lankan foods. Sri Lankan meals usually contain rice with several curries and ingredients, which makes it difficult for existing systems to correctly identify foods and estimate their nutrition.

Second, many nutrition systems still depend on manual food logging. Users need to search for food items and estimate portion sizes themselves. This process takes time and may lead to mistakes, which can reduce the accuracy of the data.

Third, many AI diet recommendation systems mainly focus on providing nutritionally balanced meals but do not consider food costs. However, affordability is an important factor for many users in developing countries.

Fourth, many existing systems use global food databases that do not fully include local ingredients and traditional Sri Lankan meals. Because of this, the nutritional information may not always be accurate.

Fifth, many applications give general diet suggestions but do not always consider the user's lifestyle, local food availability, and cultural eating habits.

To solve these problems, this research proposes **NutriAI**, an AI-powered Sri Lankan food recognition and nutrition management system. The system will combine automatic food recognition, a Sri Lankan nutrition database, and budget-friendly diet recommendations within the Aegis Life platform. This will help provide a more practical and useful nutrition management solution for people in Sri Lanka.

2. OBJECTIVE

2.1 Main Objective

To design and develop an AI-based nutrition management system that can identify Sri Lankan foods from images and provide personalized, culturally suitable, and affordable diet recommendations.

2.2 Specific Objectives

1. **To collect and label images** of common Sri Lankan foods to create a dataset for training the AI model.
2. **To build a food recognition model** using a Convolutional Neural Network (CNN) and improve its accuracy using methods like transfer learning and data augmentation.
3. **To create a Sri Lankan nutrition database** that includes calorie, nutrient, and cost information for local meals.
4. **To develop a system** that calculates users' nutrient intake and provides personalized diet advice based on their profile, health goals, and activity level.
5. **To design a meal planning algorithm** that suggests healthy and affordable meal plans according to the user's budget.
6. **To integrate NutriAI** into the Aegis Life platform so it can work together with other wellness features.

3. METHODOLOGY

This research uses an experimental and practical approach to develop a working prototype called **NutriAI** and test how well it works in real-life situations. The study follows the **Design Science Research methodology**, which focuses on designing, building, and evaluating new technological solutions. In this project, the solution is an **AI-based system that can recognize Sri Lankan food and provide nutrition management support**. The system will be tested to measure its **accuracy, reliability, and overall usability**.

For the technical part of the system, **image-based food recognition** will be used. This will be implemented using **Convolutional Neural Networks (CNNs)**, which are commonly used for image classification tasks. Pre-trained models such as **MobileNet, ResNet, and EfficientNet** will be considered because they are known for their strong performance in recognizing images. To further improve the model's accuracy, **transfer learning** will be used along with **data augmentation techniques** such as rotating images, flipping them, scaling them, and adjusting brightness.

After the system identifies a food item from an image, the detected food will be matched with a **local nutrition database** that contains nutritional information about Sri Lankan meals. The system will then estimate the **calories and nutrient values** of the meal. These values will be compared with the **recommended dietary allowances (RDA)** to provide personalized nutrition suggestions for the user. In addition, the system will include a **budget-aware meal recommendation feature** that suggests affordable and nutritionally balanced meals while also avoiding repeated meal suggestions.

The system will be developed using **Python** as the main programming language. Deep learning frameworks such as **TensorFlow or PyTorch** will be used to build and train the AI model. **OpenCV** will be used for image processing tasks. The backend of the system will be implemented using **RESTful APIs** built with **Flask or FastAPI**, which will allow communication between the AI model, database, and user interface.

The training data will come from both **publicly available datasets and locally collected images**. Public food image datasets such as **Food-101 [5]** and **UECFood256 [6]** will be used for initial model training and transfer learning. In addition, a **custom dataset of Sri Lankan foods** will be created by collecting images of common meals such as **rice and curry, hoppers, string hoppers, and kottu** through controlled photography sessions and publicly available online sources. All datasets used in this research will be properly referenced in the reference list.

To measure the performance of the system, several evaluation methods will be used. Metrics such as **accuracy, precision, recall, and F1-score** will be used to evaluate the food recognition model. The system will also measure **nutrient estimation error** and **budget compliance** to evaluate how effective the meal recommendation system is. In addition, **user feedback surveys** will be conducted to understand the system's **usability, user satisfaction, and overall usefulness**.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

Overview:

NutriAI is built in layers and works step by step. Users take a photo of their meal, and the system identifies the food, calculates nutrition, and gives personalized, budget-friendly meal advice. It is connected to the Aegis Life platform so users can see their overall health and wellness.

System Components:

- **Presentation Layer:** Where users upload meal photos and see nutrition info and recommendations.
- **Application Layer:** Manages user accounts and applies rules for diet goals and budgets.
- **AI Layer:** The "brain" of the system, which recognizes foods and estimates portion sizes.
- **Data Layer:** Stores user profiles, meal images, nutrition data, and history.

How it Works:

1. User uploads a meal photo through the app.
2. The image is securely sent to the backend server.
3. The AI model identifies the foods and estimates portion sizes.
4. The system looks up nutritional values and costs from the local database.
5. Personalized recommendations are created based on the user's health goals and budget.
6. Results are sent back to the app for the user to see.
7. All data is stored for future tracking and progress analysis.
8. The system can be expanded later for special diets or health conditions.


NutriAI System Architecture
Image Input
NutriAI
Presentation Layer
User Interface (Meal Upload, Results)
Image Analysis Request
Application Layer
User Authentication & Requests
Dietary Goals & Budget Logic
Food Data Request
AI Processing Layer
Food Recognition Model
Portion Estimation
Nutrient Analysis
Food & Nutrition Data
Data Layer
User Profiles
& Meal History
Nutrition Database

Figure 1

5. USER REQUIREMENTS

5.1 Stakeholders:

- Primary Users: University students, working adults, and the general Sri Lankan population who want nutrition guidance.
- Secondary Users: Nutritionists, dieticians, and healthcare professionals.
- System Administrators: Manage the platform, retrain AI models, and maintain the database.

5.2 Functional Requirements:

Functional requirements explain what the NutriAI system should do. These functions describe how users interact with the system and how the system processes information.

Functional Requirements

1. Upload Meal Images
   The system should allow users to upload photos of their meals using the mobile or web application.
2. Recognize Sri Lankan Foods Automatically
   The system should use an AI model to identify Sri Lankan food items from the uploaded images.
3. Calculate Nutrition Information
   The system should calculate calories and nutrients by matching the recognized foods with a Sri Lankan nutrition database.
4. Provide Personalized Diet Advice
   The system should analyze user details such as age, weight, activity level, and health goals to provide personalized nutrition advice.
5. Suggest Budget-Friendly Meals
   The system should recommend meal options that are both healthy and affordable for the user.
6. Store Meal History
   The system should save meal images, nutrition results, and dietary records securely for future reference.
7. Integrate with Aegis Life Platform
   The NutriAI module should work together with other Aegis Life features such as fitness tracking and health monitoring.

RCI (Requirement–Component Interaction)

Requirement	User Interface	Application Logic	AI Model	Database
Upload meal images	✓	✓		✓
Food recognition		✓	✓	
Nutrition calculation		✓	✓	✓
Personalized advice		✓		✓
Budget meal planning		✓		✓
Store meal history		✓		✓
System integration	✓	✓		✓

Table 2

5.3 Non-Functional Requirements:

- High accuracy in food recognition (above 80%).
- Fast response time (less than 5 seconds per image).
- Ability to support many users at the same time.
- Secure data storage and encrypted transfers.
- User-friendly interface suitable for all skill levels.
- Flexible system to add new foods or update the nutrition database easily.

6. COMMERCIALIZATION PLAN

6.1. Target Market and Customer Persona:

The primary target market for Aegis Life includes Sri Lankan adults aged 18–45 who are interested in improving their health through better nutrition and fitness tracking. This includes university students, working professionals, and health-conscious individuals. Secondary users include nutritionists, healthcare professionals, and corporate wellness programs.

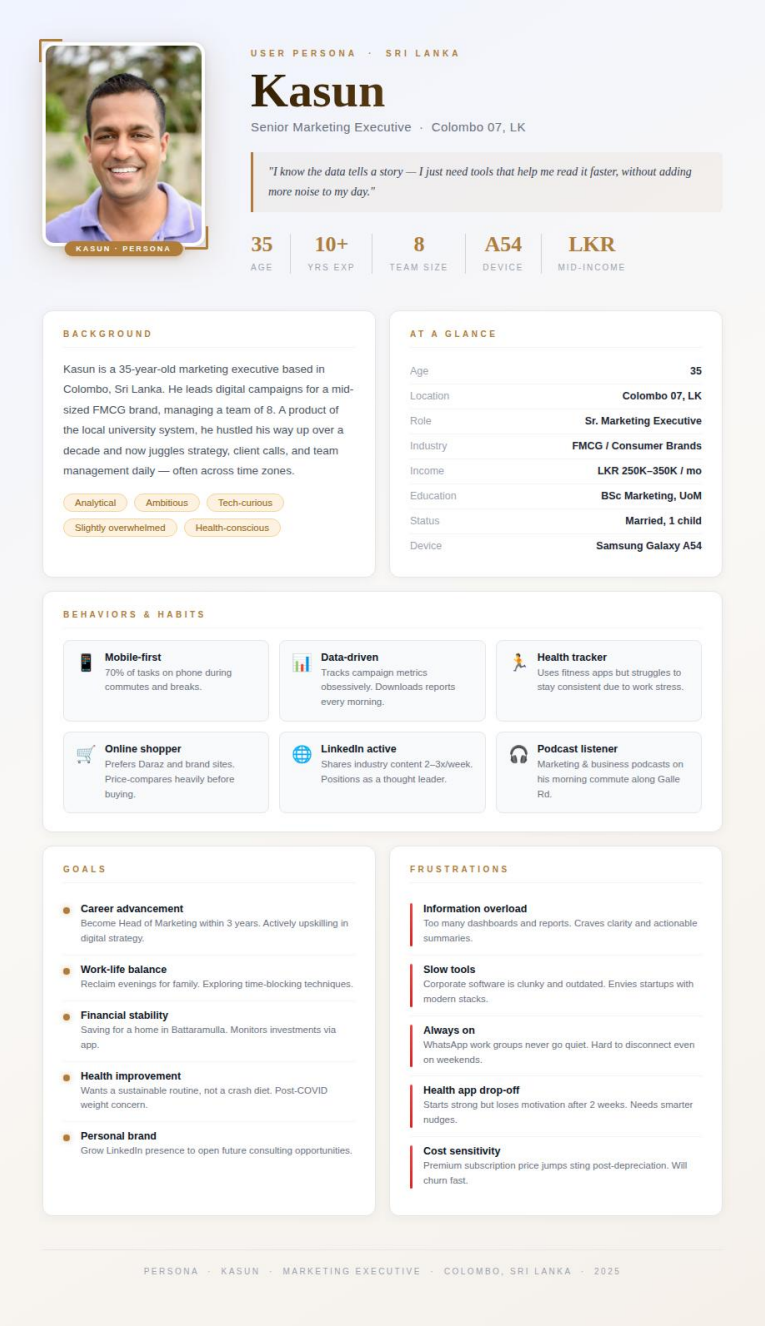


Figure 2

6.2. Value Proposition:

Aegis Life provides an integrated digital wellness platform that combines nutrition tracking, fitness monitoring, health analytics, and mental wellness support. Unlike existing applications that focus on only one aspect of health, Aegis Life connects multiple modules into a single platform to provide holistic wellness support.

6.3. Pricing Strategy:

Tier	Price	Target Users	Features
Free	Rs.0	Students and budget-conscious users	Basic workouts, manual logging, limited chatbot
Premium Monthly	Rs.599	Individual professionals	Full features including AI food recognition
Premium Annual	Rs.5,999	Long-term users	All premium features + 2 months free
Corporate	Rs.400/user/month	Companies (min 50 users)	Premium features + admin dashboard

Table 3

6.4. Cost Estimation (Annual):

Cost Category	Amount (LKR)
Cloud Services (Firebase, Hosting)	18,000
Developer Maintenance	12,000
Marketing & User Acquisition	8,000
API & Third-Party Services	6,000
Customer Support	6,000
Total Annual Cost	50,000

Table 4

6.5. Competitive Advantage:

Advantage	Aegis Life	Competitors
Integration	All 4 modules connected	Separate apps
Language	Sinhala & Tamil support	Mostly English
Wearable-Free	Works without expensive devices	Requires wearables
Local Context	Sri Lankan foods and lifestyle	Western content
Price	Rs.599/month	Higher subscription costs

Table 5

6.6. Intellectual Property (IP):

The project will protect its intellectual property through multiple mechanisms. The name **Aegis Life** and its logo will be registered as trademarks. Source code, UI/UX designs, and original content will be protected under copyright law. Novel algorithms related to multi-module health adaptation and context-aware exercise form correction may be considered for patent protection.

7. BUDGET AND JUSTIFICATION

Category	**Amount (LKR)**
Hardware Costs	31,700
Software Licenses	44,800
Cloud Services	46,500
Human Effort	18,500
Contingency	5,200
TOTAL	**146,700**

Table 6

Budget Justification

The total estimated budget for developing the Aegis Life platform is **LKR 146,700**.

Hardware costs include external storage devices and Android devices required for application testing and dataset storage. Software licenses include GitHub Pro for collaborative version control, Figma for UI design, and Expo EAS build services for mobile deployment.

Cloud services represent the largest cost because Firebase hosting, real-time databases, and cloud storage are required to support AI services and application scalability.

Human effort includes small development stipends during intensive project phases, while contingency funds are reserved for unexpected technical expenses.

8. WORK BREAKDOWN STRUCTURE (WBS)

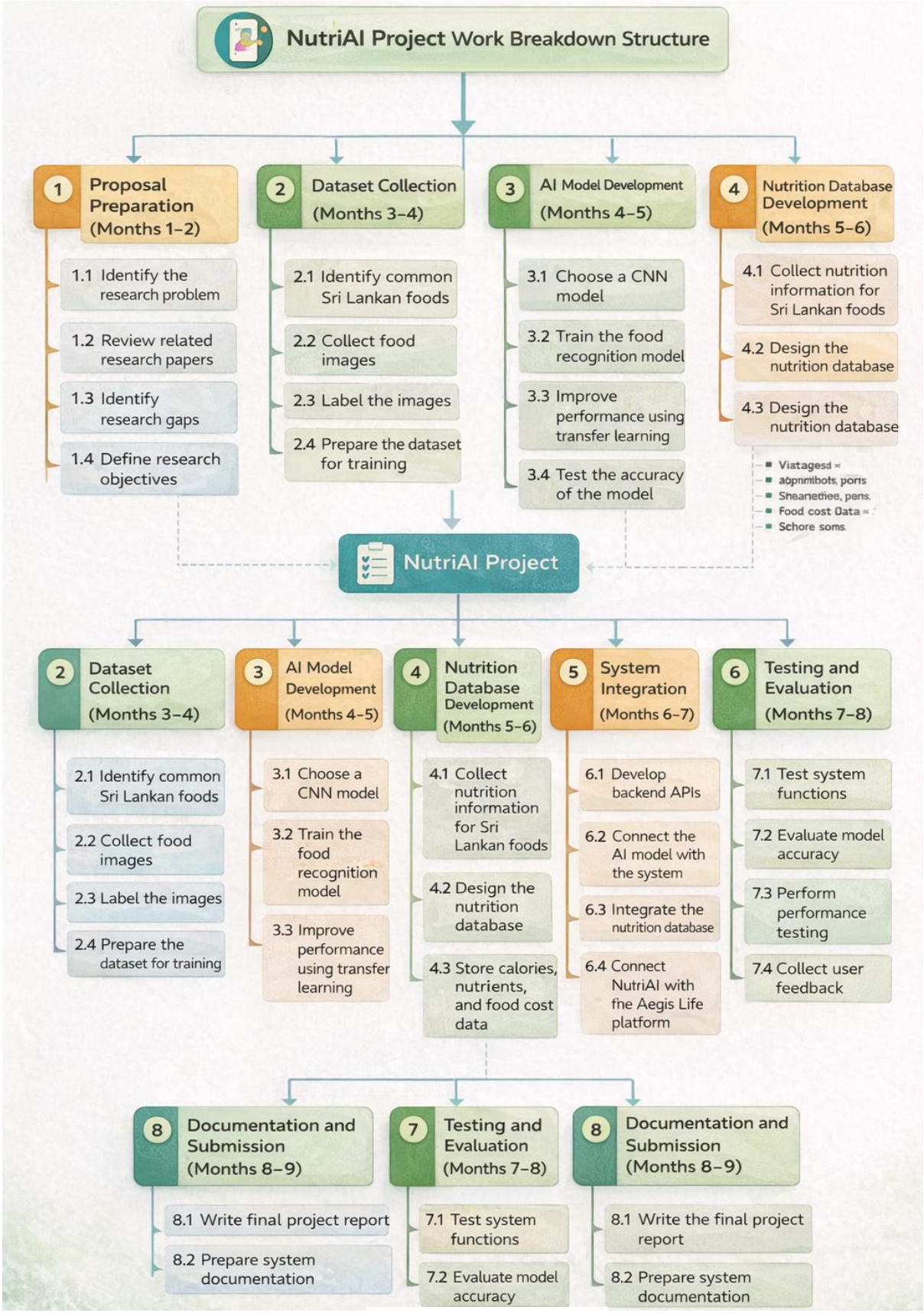


Figure 3

9. GANTT CHART

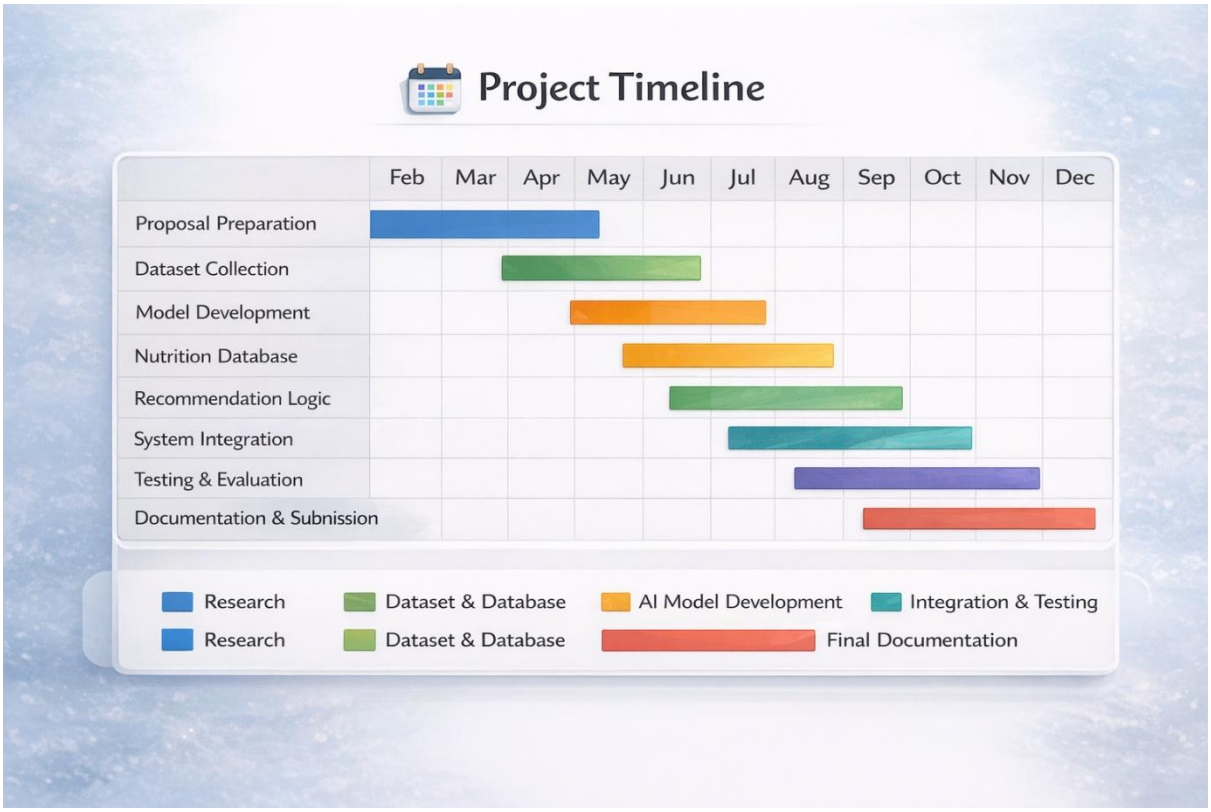


Figure 3

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11.APPENDICES

Appendix A – Sample Food Images:
Photos of Rice & Curry, Hoppers, String Hoppers, Kottu, and traditional sweets.

Appendix B – Nutrition Database Schema:
Tables with Food Name, Calories, Macronutrients, Micronutrients, and Estimated Cost.

Appendix C – User Evaluation Questionnaire:
Survey using Likert scale for usability, satisfaction, usefulness, and reliability.

Appendix D – System Architecture Diagram:
Layered diagram: Presentation Layer → Application Layer → AI Processing Layer → Data Layer.

Appendix E – Risk Analysis:

Risk	Impact	Mitigation
Incorrect classification	High	Model retraining, confidence thresholds
Inaccurate nutrient estimation	Medium	Validate database with references
Data breaches	High	Encryption and access control
User disengagement	Medium	Gamification, reminders

Table 7

Appendix F – UI Mockups:
Screens for meal image upload, recognition results, nutrient summary, and budget-aware recommendations.